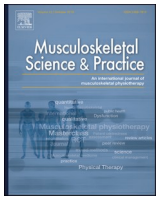
Musculoskeletal Science and Practice 82 (2026) 103525



Contents lists available at ScienceDirect

Musculoskeletal Science and Practice

journal homepage: www.elsevier.com/locate/msksp





International normative values for the weight-bearing lunge test across age and sex in 899 healthy adults

Sean McBride [a], Pier-Luc Isabelle [b], Álvaro Gómez-Carrión [c], Nader Farahpour [d], Dominic Chicoine [b], Ian Griffiths [e], Ahmed Dami [f], Jose Manuel Reguera Medina [g], Gabriel Moisan [b,f,*]

[a] Department of Rehabilitation Sciences, Medical University of South Carolina, 151-B Rutledge Avenue, MSC 339, Charleston, SC, 29425, USA
[b] Department of Human Kinetics, Université Du Québec à Trois-Rivières, 3351, Boul. des Forges, C.P.500, Trois-Rivières, G8Z 4M3, Canada
[c] Department of Nursing and Podiatry, University of Malaga, Avda. Cervantes, 2, Málaga, 29071, Spain
[d] Department of Sport Biomechanics, Faculty of Sport Sciences, Bu-Ali Sina University, QFWQ+C8V District 1, Hamedan, Iran
[e] Sports and Exercise Medicine, William Harvey Research Institute, Queen Mary University of London, Mile End Hospital, Bancroft Road, London, E1 4DG, United Kingdom
[f] Groupe de Recherche sur les Affections Neuromusculosquelettiques (GRAN), Université Du Québec à Trois-Rivières, 3351, Boul. des Forges, C.P.500, Trois-Rivières, G8Z 4M3, Canada
[g] Sanipie Clinic, C. La Monteria, 27, Utrera, 41710, Sevilla, Spain

ARTICLE INFO

Keywords:
Foot
Ankle
Physical examination
Biomechanical phenomena
Podiatry

ABSTRACT

Background: Ankle dorsiflexion is essential for lower extremity function and limitations in this motion are associated with musculoskeletal injury and decreased performance during functional tasks. The distance-based weight-bearing lunge test (WBLT) is a reliable and valid clinical measure of ankle dorsiflexion, yet normative data across age and sex are lacking.
Objectives: To establish adult age- and sex-specific normative values for the distance-based WBLT.
Design: Cross-sectional, multi-country observational study.
Method: A total of 899 adults from Canada, Spain, and Iran completed demographic and clinical measures including height, Foot Posture Index (FPI-6), supination resistance, and bilateral WBLT performance using a standardized knee-to-wall protocol. Participants were stratified into seven age groups and by sex. Between-limb differences were examined with paired t-tests. Associations between WBLT and height, FPI-6, and supination resistance were assessed using Pearson correlations. Age and sex effects were evaluated using two-way ANOVA with Bonferroni-adjusted comparisons. Percentile-based clinical categories were generated for each age–sex subgroup.
Results/findings: No meaningful difference was observed between limbs ($p = 0.172$, $d = 0.05$). WBLT performance declined progressively with age ($p < 0.001$, $\eta^2 = 0.283$), with greater reductions evident beginning in the 60-69 age group. Males demonstrated slightly greater dorsiflexion than females across age groups ($p = 0.003$, $\eta^2 = 0.010$). Height, FPI-6, and supination resistance showed small, non-clinically meaningful associations with WBLT scores ($R^2 \leq 0.09$).
Conclusions: This study provides robust age- and sex-specific normative values for the distance-based WBLT, improving clinical interpretation of weight-bearing ankle dorsiflexion across the adult lifespan.

1. Introduction

Sufficient ankle dorsiflexion range of motion is required for normal function. This motion allows forward progression of the tibia during gait, enables proper shock absorption, and supports efficient lower-extremity alignment during closed-chain tasks (Neumann, 2017;

* Corresponding author. Université du Québec à Trois-Rivières, Canada.
E-mail addresses: mcbridse@musc.edu (S. McBride), pier-luc.isabelle@uqtr.ca (P.-L. Isabelle), alvgom25@uma.es (Á. Gómez-Carrión), naderfarahpour1@gmail.com (N. Farahpour), dominic.chicoine@uqtr.ca (D. Chicoine), i.griffiths@qmul.ac.uk (I. Griffiths), ahmed.dami@uqtr.ca (A. Dami), josemaremed22@gmail.com (J.M. Reguera Medina), Gabriel.moisan@uqtr.ca (G. Moisan).

https://doi.org/10.1016/j.msksp.2026.103525
Received 5 January 2026; Received in revised form 17 February 2026; Accepted 19 February 2026
Available online 20 February 2026
2468-7812/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Almansoof et al., 2023). Reduced dorsiflexion has been associated with several lower-extremity pathologies, including plantar fasciopathy (Riddle et al., 2003), inversion ankle sprains (Willems et al., 2005), and patellar tendinopathy (Backman and Danielson, 2011; Malliaras et al., 2006). Functional activities, such as walking and running (Rao et al., 2024), negotiating stairs (Protopapadaki et al., 2007), squatting (Dill et al., 2014), and landing from a jump (Han et al., 2023) are also negatively affected by decreased dorsiflexion. Individuals with reduced dorsiflexion measured during a static clinical test also demonstrated altered gait mechanics, including reduced dorsiflexion during gait and a shorter step length (Aquino et al., 2022). Additionally, the dorsiflexion range of motion declines across the lifespan (Vandervoort et al., 1992; Soucie et al., 2011), which may potentially impair dynamic balance and increase fall risk (Kemoun et al., 2002; Hernández-Guillén et al., 2021). Collectively, these consequences highlight the importance of accurate clinical measurement to determine whether an individual demonstrates normal ankle dorsiflexion range of motion.

Clinicians use various methods to measure ankle dorsiflexion, including both non–weight-bearing (NWB) and weight-bearing (WB) assessments. WB measures, such as the distance-based weight-bearing lunge test (WBLT), which quantifies maximal toe-to-wall distance, capture dorsiflexion in a functional closed-chain position, with alternative angle-based quantification methods also described in the literature. Compared to NWB methods, the WBLT has revealed larger between-limb differences not apparent during NWB assessment (Koshino et al., 2024).

For a clinical test to be useful, it must demonstrate strong reliability and validity, and its results must be interpretable against relevant reference data. The distance-based WBLT demonstrates excellent reliability, with pooled interrater intraclass correlation coefficient (ICC) values averaging 0.95 (SE = 0.02) and similarly high intrarater reliability (ICC = 0.93, SE = 0.03) (Powden et al., 2015). The excellent reliability of the test has also been extended to individuals with various pathologies (Gosse et al., 2021; Simondson et al., 2012, 2022). Weight-bearing dorsiflexion measured during the lunge test has also been shown to reflect talocrural joint dorsiflexion when compared with radiographic assessment, supporting its validity as a clinical measure (Smith et al., 2019). However, the clinical interpretability of an individual's score is hindered by a lack of robust normative data. Existing studies (Nambiar et al., 2024; Gohil and Tilaye, 2022; Onate et al., 2018) are often limited by small sample sizes and crucially fail to provide reference values stratified by key factors such as age or sex. While normative data for the angle-based WBLT exist (McKay et al., 2017), normative data for the distance-based WBLT are lacking. The distance-based test demonstrates greater reliability (Powden et al., 2015), does not require an inclinometer, and may provide a more global foot and ankle dorsiflexion perspective compared to the angle-based measure. However, clinicians lack reference ranges to determine whether an individual's dorsiflexion is within normal limits or represents a meaningful restriction.

Therefore, the purpose of this study was to establish age- and sex-stratified normative distance-based WBLT dorsiflexion values in a large international sample of healthy adults across age and sex groups.

2. Methods

2.1. Participants

To establish normative values, at least 50 participants per group are recommended to reduce variability (Bridges and Holler, 2007). Given that this study includes seven age groups and two sexes per age group (total of 14 subgroups), a minimum of 700 participants was required. We recruited participants until this minimum was exceeded and a robust, feasible international sample was obtained. A total of 899 individuals were recruited between January 2024 and October 2025 from Canada (n = 403), Spain (n = 344) and Iran (n = 152) to participate in this international multicenter study. In Canada, participants were recruited from the outpatient podiatry clinic, and the staff and students from the Université du Québec à Trois-Rivières. Additionally, data were collected during two provincial conferences: the Quebec College of Podiatry Conference held in June 2024, and the Integrative Health Conference held in January 2025, both in Saint-Hyacinthe, Québec, Canada. Participants in Iran were recruited from the Exercise centre specialised in spine care (Mehr Corporation, Hamedan, Iran), the student and staff population of Bu-ali Sina University, and local recreational facilities. In Spain, recruitment took place at the biomechanics and podiatry unit of San Agustín Hospital (Dos hermanas, Sevilla, Spain) and from the Sanipie Clinic podiatry centre, (Utrera, Seville, Spain).

To be included in the study, all participants needed to be 18 years or older. The exclusion criteria were as follows: experiencing a lower limb musculoskeletal disorder within 3-month prior to data collection, having a history of a lower limb orthopedic surgery, being diagnosed with a neuromuscular disorder that could alter balance control or foot and ankle morphology (e.g., multiple sclerosis, Parkinson's disease, stroke), or ongoing pregnancy.

The study protocol was approved by the Université du Québec à Trois-Rivières Ethic Committee (CER-23-305-07; Canada, Iran), and Bioética y Bioseguridad de Universidad de Extremadura (77/2024, Spain). All participants provided written informed consent prior to the experimental procedures.

3. Protocol

Demographic data, consisting of age, sex, body mass, height, supination resistance (Moisan et al., 2022), foot posture assessed with the Foot Posture Index (FPI) (Redmond et al., 2006), and WBLT scores were recorded (see Table 1).

3.1. Weight-bearing lunge test

Weight-bearing lunge test data were collected for both feet following a standardized knee-to-wall protocol using a measuring tape placed on the floor (see Fig. 1). (Bennell et al., 1998) Participants were instructed to stand facing a wall without a baseboard with the tested foot positioned perpendicular to it and the opposite foot placed comfortably behind for balance. Keeping the heel of the tested foot flat on the ground, participants were asked to slowly lunge forward, attempting to touch their knee to the wall without lifting the heel. If the heel lifted, the foot was repositioned slightly closer to the wall. If the knee touched the wall while maintaining full heel contact, the foot was moved gradually backward until the maximal distance allowing heel contact and knee-wall touch was achieved. The distance from the tip of the hallux to the wall was then measured to the nearest millimeter using the floor-mounted measuring tape. Participants were allowed to use the wall in front of them as support if needed.

3.2. Statistical analysis

Data were analyzed with SPSS-29.0.2.0 (IBM Corporation, Armonk, NY, USA). WBLT data were averaged according to age groups: 18 to 29, 30 to 39, 40 to 49, 50 to 59, 60 to 69, 70 to 79, and 80+ years. WBLT data were also disaggregated according to sex for a total of 14 subgroups. Then, the normality of the distribution of the demographic and WBLT data was evaluated using visual (histograms/probability graphs) methods.

Given the normal distribution of the WBLT data, pooled data across age groups and sexes was compared between the right and left foot using a paired *t*-test. Considering the large sample size, the level of statistical significance was set at $\alpha = 0.050$ and $d \geq 0.50$ for this analysis. Because no meaningful side-to-side difference was observed, a single value per participant was used. In studies involving paired limbs, the person rather than the limb is considered the unit of analysis, and a

Table 1
Demographic data.

Age group (yrs)	Sex	n	Age (yrs)[a]	Mass (kg)[a]	Height (m)[a]	FPI-6[a]	WBLT (cm)[a]
18–29	Male	51	25.5 (2.6)	84.1 (16.4)	1.77 (0.08)	4.6 (3.6)	11.0 (3.6)
	Female	90	23.6 (2.7)	65.1 (12.7)	1.63 (0.07)	4.8 (3.1)	11.3 (3.0)
30–39	Male	59	34.5 (3.2)	86.1 (16.3)	1.78 (0.08)	4.0 (3.4)	11.2 (3.8)
	Female	79	34.2 (2.8)	67.0 (12.4)	1.64 (0.06)	3.0 (2.7)	11.4 (3.0)
40–49	Male	55	44.9 (2.8)	84.0 (12.5)	1.77 (0.07)	4.1 (3.0)	11.0 (3.2)
	Female	97	43.7 (2.5)	71.2 (13.8)	1.63 (0.06)	3.1 (3.2)	10.4 (2.7)
50–59	Male	62	53.5 (2.5)	83.2 (11.9)	1.74 (0.08)	4.0 (3.9)	10.4 (3.6)
	Female	78	54.5 (2.9)	72.1 (16.8)	1.63 (0.06)	3.3 (3.1)	9.5 (2.6)
60–69	Male	50	64.3 (3.0)	87.4 (16.3)	1.73 (0.06)	3.9 (3.1)	9.5 (3.3)
	Female	73	63.5 (2.5)	72.0 (16.3)	1.60 (0.07)	3.8 (3.0)	8.0 (2.6)
70–79	Male	53	75.0 (2.6)	83.3 (14.1)	1.68 (0.07)	4.7 (3.6)	7.4 (2.3)
	Female	52	74.5 (3.1)	71.1 (13.5)	1.57 (0.06)	5.5 (3.2)	6.5 (2.2)
80+	Male	50	83.7 (4.1)	81.7 (13.4)	1.68 (0.07)	4.3 (3.1)	6.6 (2.0)
	Female	50	84.2 (4.0)	70.7 (12.3)	1.53 (0.08)	5.4 (2.7)	5.7 (1.7)
All groups	Male	380	53.9 (19.5)	84.3 (14.5)	1.74 (0.08)	4.2 (3.4)	9.6 (3.6)
	Female	519	50.2 (19.5)	69.7 (14.3)	1.61 (0.07)	4.0 (3.2)	9.4 (3.3)

Abbreviations: yrs = years; n = number; kg = kilograms; m = meters; FPI-6 = six-item Foot Posture Index; WBLT = Weight-bearing lunge test; cm = centimeters.
[a] Mean (Standard Deviation).

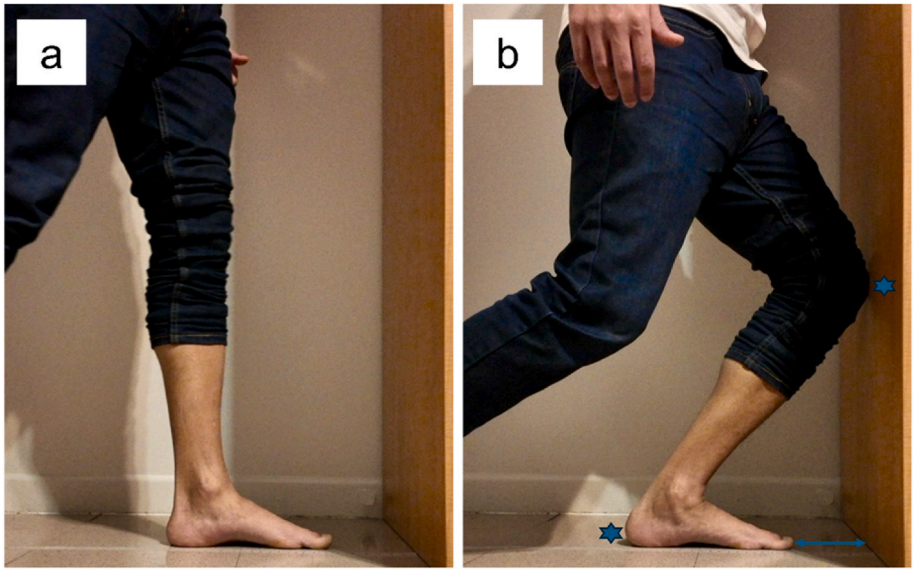


Fig. 1. The starting position (a) and the final position (b) for the distance-based weight-bearing lunge test. The * indicates that the heel and knee must be in contact with the floor and wall, respectively. The line with arrows indicates the distance measurement that is recorded as the test score.

representative limb may be selected (Menz, 2005). We reported the dominant foot because the WBLT is interpreted clinically on an individual limb basis. We did not average both limbs since this would create a derived value that is not directly used in clinical decision-making. The dominant foot was determined to be the one that a participant would select if asked to kick a ball.

Bivariate regression analyses between WBLT and height, between WBLT and supination resistance, and between WBLT and the FPI-6 score were conducted with Pearson's correlations (r) and 95% confidence intervals to identify potential covariates. The coefficient of determination (R^2) was also calculated. Considering the large sample size, the level of statistical significance was set at $\alpha = 0.050$ and $r \geq 0.40$ (Schober et al., 2018).

Considering that no significant covariate was identified, a two-way ANOVA was performed with "age group" (seven levels) and "sex" (two levels) as independent factors. WBLT scores were compared using Bonferroni-adjusted pairwise comparisons for the main effects of age group and sex, as well as for their interaction. The significance level was set at $\alpha = 0.050$ for these analyses. Clinical categories based on percentiles, stratified by age group and sex, were calculated for the WBLT: very low (<2.3), low (2.3-15.9), slightly low (15.9-25.0), normal (25.0-75.0), slightly high (75.0-84.1), high (84.1-97.7) and very high (>97.7).

4. Results

There were no significant differences between WBLT scores for the left and right foot (9.5 ± 3.5 vs. 9.5 ± 3.4 cm, $p = 0.172$, $d = 0.05$). There were no significant correlations between height ($r = 0.301$ (95% CI: 0.240-0.359), $p < 0.001$, $R^2 = 0.09$), FPI-6 ($r = -0.116$ (95%CI: −0.180—-0.051), $p < 0.001$, $R^2 = 0.01$), supination resistance ($r = 0.245$ (95%CI: 0.183-0.306), $p < 0.001$, $R^2 = 0.06$) and WBLT scores (see Fig. 2 and Table 2).

4.1. Age group effects

There was a significant age group effect ($F = 58.3$, $p < 0.001$, $\eta 2 = 0.283$). Regardless of sex, WBLT score in the 80+ age group was smaller than in the 60-69 age group (−2.6 cm, $p < 0.001$), 50-59 (−3.8, $p < 0.001$), 40-49 (−4.6 cm, $p < 0.001$), 30-39 (−5.2 cm, $p < 0.001$) and 18-29 (−5.0 cm, $p < 0.001$) age groups. WBLT score in the 70-79 age group

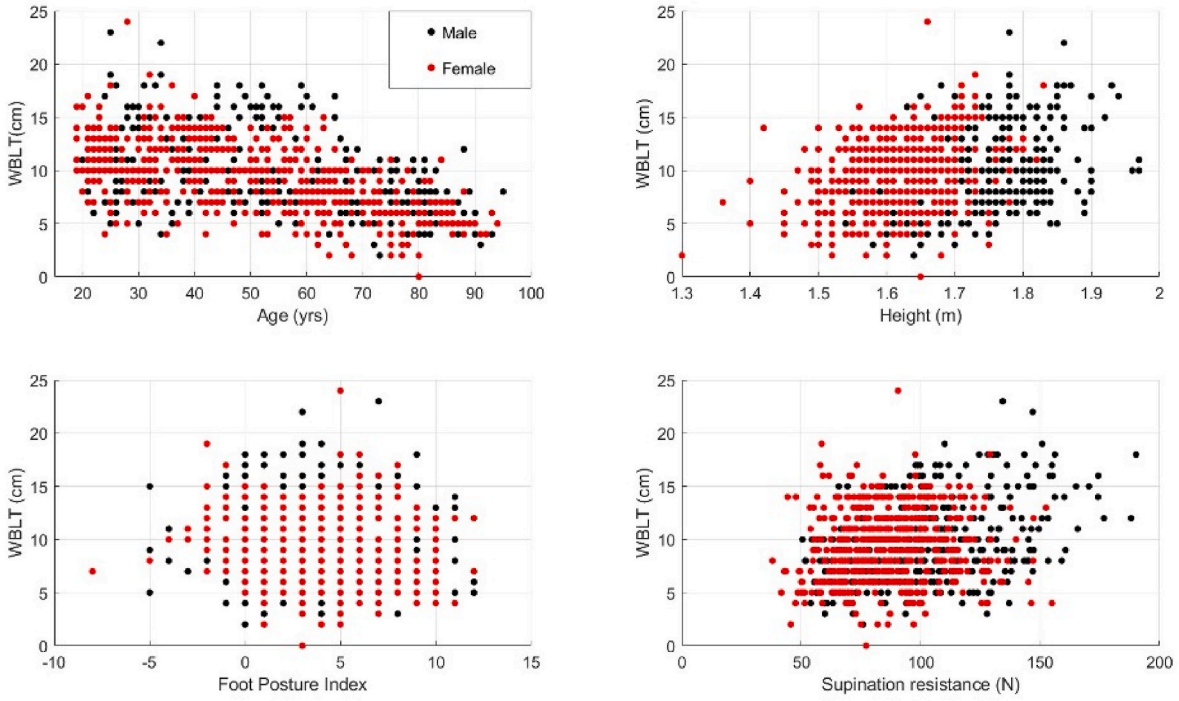


Fig. 2. Scatterplots of WBLT scores in relation to age (top left), height (top right), Foot Posture Index (bottom left) and supination resistance (bottom right).

Table 2
Clinical categories for the weight-bearing lunge test according to age and sex.

Age Group	Sex	Clinical Categories for the Distance-based Weight-bearing Lunge Test (cm)						
		Very low	Low	Slightly low	Normal	Slightly high	High	Very high
18-29	Male	<4.7	4.7-7.2	7.2-8.0	8.0-13.2	13.2-14.2	14.2-22.1	>22.1
	Female	<4.7	4.7-9.0	9.0-9.8	9.8-12.8	12.8-13.7	13.7-17.9	>17.9
30–39	Male	<4.3	4.3–7.3	7.3–8.8	8.8–14.0	14.0–15.1	15.1–20.5	>20.5
	Female	<4.9	4.9–7.6	7.6–9.8	9.8–13.8	13.8–14.0	14.0–17.9	>17.9
40-49	Male	<6.1	6.1-7.7	7.7-8.0	8.0-14.0	14.0-14.8	14.8-18.0	>18.0
	Female	<4.6	4.6-7.8	7.8-8.6	8.6-13.0	13.0-13.5	13.5-14.5	>14.5
50–59	Male	<4.9	4.9–7.0	7.0–7.5	7.5–13.8	13.8–15.5	15.5–17.8	>17.8
	Female	<4.0	4.0–7.0	7.0–7.5	7.5–11.5	11.5–11.9	11.9–14.5	>14.5
60-69	Male	<4.2	4.2-6.4	6.4-7.0	7.0-10.6	10.6-12.5	12.5-17.2	>17.2
	Female	<2.2	2.2-5.8	5.8-7.0	7.0-9.0	9.0-10.1	10.1-13.8	>13.8
70–79	Male	<2.0	2.0–5.0	5.0–5.8	5.8–9.9	9.9–10.0	10.0–11.0	>11.0
	Female	<2.0	2.0–4.0	4.0–5.0	5.0–8.0	8.0–9.0	9.0–10.8	>10.8
80+	Male	<3.2	3.2-4.8	4.8-5.0	5.0-8.0	8.0-8.4	8.4-11.8	>11.8
	Female	<0.3	0.3-4.7	4.7-5.0	5.0-6.2	6.2-7.0	7.0-10.8	>10.8

was smaller than in the 60-69 (−1.8 cm, $p < 0.001$), 50-59 (−3.0 cm, $p < 0.001$), 40-49 (−3.7 cm, $p < 0.001$), 30-39 (−4.4 cm, $p < 0.001$) and 18-29 (−4.2 cm, $p < 0.001$) age groups. WBLT score in the 60-69 age group was smaller than in the 50-59 (−1.2 cm, $p = 0.017$), 40-49 (−2.0 cm, $p < 0.001$), 30-39 (−2.6 cm, $p < 0.001$) and 18-29 (−2.4 cm, $p < 0.001$) age groups. WBLT score in the 50-59 age group was smaller than in the 30-39 (−1.4 cm, $p = 0.002$) and 18-29 (−1.2 cm, $p = 0.016$) age groups (see Fig. 3).

4.2. Sex effects

Of the 899 participants, 380 were male (42.3%) and 519 were female (57.7%). There was a significant sex effect ($F = 9.07$, $p = 0.003$, $\eta 2 = 0.010$). Regardless of the age group, male participants exhibited greater WBLT scores than female participants (9.6 vs 9.0 cm, $p = 0.003$).

4.3. Interaction effects

No significant interaction effect was observed ($F = 1.56$, $p = 0.157$, $\eta 2 = 0.010$).

5. Discussion

This study sought to establish age- and sex-specific normative values for the distance-based WBLT in a large international adult sample ($n = 899$). Ankle dorsiflexion range of motion declined progressively with age, with more pronounced reductions emerging beginning with the 60-69 age group and continuing into the 70-79 and 80+ age groups. Men demonstrated slightly greater dorsiflexion than women (with a small effect size), and no meaningful between-foot differences were observed within participants. Correlations between WBLT performance and height, foot posture (FPI-6), and supination resistance were small and not clinically relevant.

To the authors' knowledge, this is the first study to present normative data from a large, international sample of adults across the lifespan for the distance-based WBLT. McKay et al. (2017) presented normative data for the angular measurement of the WBLT across 1000 Australian individuals from 3 to 101 years. Values between these studies cannot be

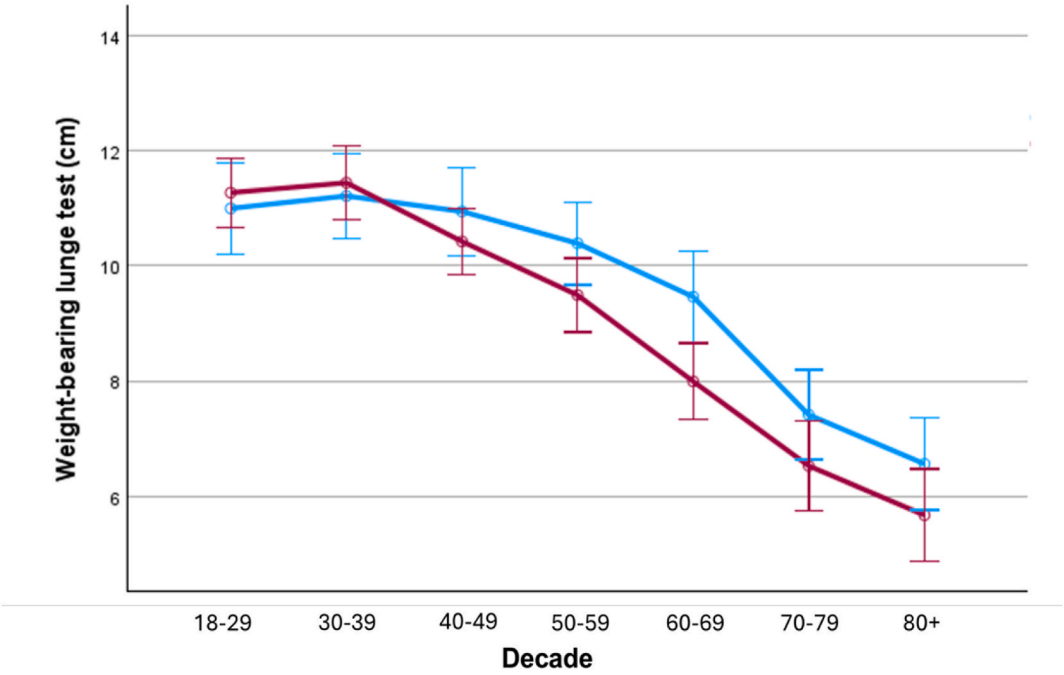


Fig. 3. Weight-bearing lunge test scores related to age and sex. Blue curve: male participants, Red curve: female participants. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

directly compared due to the difference in measurement technique. Other normative data for the distance-based WBLT are limited and typically restricted to specific age groups or single-sex samples. Three studies have reported WBLT values in younger adults. Nambiar et al. (2024) examined 400 non-athlete females (18–26 years) and reported mean dorsiflexion distances of 11.16 ± 2.86 cm and 10.36 ± 2.62 cm for the right and left limbs, respectively. These values correspond closely with our findings for women aged 18–29 (11.3 ± 3.0 cm), particularly given that the right limb served as the dominant leg in 92.5% of their sample. Gohil and Tilaye (2022) assessed 215 men and women aged 20–30 and reported mean values of 14.39 ± 2.73 cm for males and 13.34 ± 3.37 cm for females. Although their values were higher than those observed in our sample, the means fall within one standard deviation of our corresponding age-matched groups. Notably, their protocol recorded the best of three trials, which may have increased measured dorsiflexion through repeated exposure and the potential for acute flexibility gains. Many participants in our study required several attempts for correct foot placement as well, but the number of attempts was likely overall less since we did not capture the best of three separate trials.

Onate et al. (2018) reported normative WBLT values from a very large cohort of 3951 high-school athletes aged 13–19 years. Because our study included adults only, direct comparison is challenging. Their sample demonstrated mean dorsiflexion distances of 10.1 cm for males and 10.3 cm for females, a pattern that differs from our findings and those of Gohil and Tilaye (2022), who observed greater dorsiflexion in males. The reversal of the sex difference in the Onate et al. (2018) cohort may reflect adolescent-specific factors such as musculoskeletal maturation, training exposure, or stage of physical development rather than true adult patterns of ankle mobility. These findings reinforce the importance of age-specific normative values when interpreting WBLT performance.

No meaningful between-limb difference in the WBLT was found in our study ($p = 0.172$). Hoch et al. (Hoch and McKeon, 2011) examined asymmetry in 35 healthy adults and reported a mean left–right difference of 0.1 ± 1.5 cm, with a reference range of approximately −2.7 to 2.9 cm. Although their study was not intended to provide population-wide normative values, it demonstrated that substantial asymmetry is uncommon in healthy individuals. Our much larger international sample confirmed this pattern, showing no significant difference between limbs. Taken together, these findings support the use of dominant-limb values when generating normative reference ranges. However, clinicians should still assess each limb independently to avoid overlooking meaningful dorsiflexion asymmetries in clinical populations.

Across the adult lifespan, dorsiflexion range of motion in this study declined with age, a finding that aligns with the general trends of the data from previous studies (Soucie et al., 2011; McKay et al., 2017). However, the finding in our study of the distance-based WBLT appears to demonstrate a stronger relationship with age than that observed by McKay et al. (2017) in their study on the angle-based WBLT. In McKay et al. (2017), dorsiflexion decreased by approximately 1° in males and 3° in females from the 20–59 to the 60+ age group. Conversely, in our study, mean WBLT distance decreased by 4.4 cm and 5.6 cm for males and females, respectively, from 18 to 80 years of age. In their study (McKay et al., 2017), the angle-based WBLT demonstrated a weak, inverse association with age ($r = -0.205$) whereas a moderate association was revealed in our data ($r = -0.520$). Our finding indicates that age accounted for 27% of the variation in the distance-based WBLT, compared to 4.2% for the angular method. This discrepancy may be explained by the different kinematics captured by the two different measurement techniques. The angular method captures the inclination of the shank during dorsiflexion but does not account for the anterior translation of the shank over the talus and remaining foot joints like the distance-based method. With aging, the prevalence of foot osteoarthritis at various joint increases (Roddy et al., 2015), likely contributing to increased rigidity (Menz, 2015) that may limit motion and thereby decrease the score for this test.

Several mechanisms likely contribute to the age-related decline in dorsiflexion observed in this and other studies. Structural changes within the joint capsule occur with aging, including increased collagen cross-linking and regions of fibrocartilaginous thickening (Ralphs and Benjamin, 1994), which may reduce capsular compliance and limit end-range dorsiflexion. Although tendon stiffness tends to decrease with age, muscle stiffness increases, and this shift appears to compromise overall joint mobility (Marcucci and Reggiani, 2020). Prior research has

consistently shown that older individuals exhibit less maximal passive dorsiflexion than younger adults (Giuliani et al., 2019; Gajdosik et al., 1999). In men, older adults demonstrated greater hysteresis, suggesting deterioration in muscle tissue quality (Giuliani et al., 2019), and in women, dorsiflexion was negatively associated with calf muscle–tendon unit length (Gajdosik et al., 1999). Collectively, these findings suggest that age-related changes in capsular, muscular, and muscle–tendon unit properties contribute to reductions in dorsiflexion with age.

Dorsiflexion range of motion as measured during the WBLT was generally greater in males than in females across the adult lifespan. In the 18–29 and 30–39-year age groups, mean WBLT values for females were slightly greater than those for males, but these differences were not statistically significant. From 40 years of age onward, dorsiflexion was greater in males than in females. Height accounted for approximately 9% of the variance in WBLT scores, which would tend to favor higher values in males, but this effect size is small. Vandervoort et al. (1992) showed that females displayed a greater loss of passive dorsiflexion range of motion with age compared to males. Mean dorsiflexion in females decreased from 19.3 ± 3.2° in those aged 55–60 years to 12.1 ± 5.5° in those aged 80–85 years, whereas corresponding values for males decreased from 15.4 ± 4.3 to 13.1 ± 3.5°. Soucie et al. (2011) reported a similar pattern using seated passive dorsiflexion measured with a goniometer. Mean dorsiflexion in females decreased from 13.8° in those aged 20–44 years to 11.6° in those aged 45–69 years, whereas corresponding values for males decreased from 12.7 to 11.9°, indicating a larger age-related decline in females and convergence of male and female values in older adults. Overall, these findings suggest that women experience a greater age-related loss of dorsiflexion than men, and our WBLT results extend this pattern to a functional, weight-bearing measure. From 40 to 49 years onward, males demonstrated a 40% decline in the WBLT score (from 11.0 cm to 6.6 cm), compared to a 45.2% decline in females (10.4 cm to 5.7 cm). The greatest difference in decline occurred between the 50-59 and 60-69 decades, where males declined by 8.7% compared to 15.8% for females. The accelerated loss of ankle dorsiflexion in females may be reflect multiple contributing factors, including age-related hormonal changes affecting musculoskeletal tissues (Roman-Blas et al., 2009), customary footwear differences (e.g. habitual heel use) (Csapo et al., 2010), and sex-related differences in physical activity patterns (Hawkins et al., 2009). Clinicians should be aware of the steeper age-related decline in dorsiflexion for females and interpret scores on the distance-based WBLT according to the age- and sex-specific categories presented here.

The WBLT was not related to foot posture or supination resistance, which largely reflect frontal and transverse plane posture and mobility. When considered alongside the small effect of height on WBLT performance, our findings suggest that the WBLT primarily reflects sagittal plane joint mobility.

The age- and sex-specific percentiles presented in this study can be used to guide clinicians in determining whether distance-based WBLT values fall within an expected range for an individual. These values may assist in clinical decision making and may help contextualize clinical benchmarks and rehabilitation goals. For example, the ‘Low’ and ‘Very Low’ categories represent values below the 15th percentile, indicating that these individuals display less dorsiflexion than approximately 85% of their respective age–sex cohort. When considered alongside the overall examination, findings in this range may support clinical reasoning related to mobility restriction and help guide impairment-focused interventions or monitoring of change over time. Scores within the ‘Slightly Low’ to ‘Slightly High’ categories fall between the 25th and 75th percentiles and represent the range of normal associated with the middle 50% of individuals in that group. Values in the ‘High’ and ‘Very High’ categories exceed the 85th percentile and reflect greater-than-typical dorsiflexion. When present with other features of generalized or localized hypermobility, these findings may support clinical reasoning emphasizing stability and movement control. We cannot determine from this study at which points mobility at either extreme confers benefit or risk, and the distance-based WBLT should be interpreted in combination with other examination findings. Nonetheless, these reference values provide a practical framework for identifying potentially meaningful mobility deviations and for interpreting WBLT performance across the adult lifespan.

This study has several limitations. First, the data come from a cross-sectional design, which does not allow for inferences regarding within-person changes across the age groups. Second, participants were recruited by convenience from clinical settings, universities, and professional conferences in three countries, which may not fully represent the broader adult population within each age group and may introduce selection bias toward more health-conscious or active individuals. Third, although previous work suggests excellent interrater reliability for the distance-based WBLT, we were unable to calculate the reliability of our raters due to the international nature of this study. Finally, the normative data presented here are for the distance-based WBLT and should not be assumed to translate to the angle-based WBLT.

6. Conclusion

Distance-based WBLT dorsiflexion values were slightly greater in males than in females and declined progressively with age in this large international sample of healthy adults. The age- and sex-specific normative values presented in this study provide a reference framework to assist clinicians and researchers in determining whether an individual's dorsiflexion falls within an expected range. Prospective, longitudinal investigations are needed to clarify whether above or below average of ankle dorsiflexion using these WBLT norms predisposes individuals to develop musculoskeletal disorders.

CRediT authorship contribution statement

Sean McBride: Writing – review & editing, Writing – original draft, Methodology, Funding acquisition. **Pier-Luc Isabelle:** Writing – review & editing, Funding acquisition, Data curation. **Álvaro Gómez Carrión:** Writing – review & editing, Funding acquisition, Data curation. **Nader Farahpour:** Writing – review & editing, Funding acquisition, Data curation. **Dominic Chicoine:** Writing – review & editing, Funding acquisition, Data curation. **Ian Griffiths:** Writing – review & editing, Data curation. **Ahmed Dami:** Writing – review & editing, Data curation. **Jose Manuel Reguera Medina:** Writing – review & editing, Data curation. **Gabriel Moisan:** Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Gabriel Moisan reports financial support was provided by Ordre des Podiatres du Québec. Pier-Luc Isabelle reports financial support was provided by Fondation de l’UQTR. Gabriel Moisan reports financial support was provided by Fonds de Recherche du Québec-Santé. Pier-Luc Isabelle reports financial support was provided by Syndicat des chargés de cours de l’UQTR. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors would like to thank Valerie Jomphe PhD, statistician, for her help with the statistical analyses of this study.

References

Almansoof, H.S., Nuhmani, S., Muaidi, Q., 2023. Role of ankle dorsiflexion in sports performance and injury risk: a narrative review. Electron. J. Gen. Med. 20 (5). https://doi.org/10.29333/ejgm/13412 em521.

Aquino, M.R.C., Resende, R.A., Kirkwood, R.N., Souza, T.R., Fonseca, S.T., Ocarino, J.M., 2022. Spatial-temporal parameters, pelvic and lower limb movements during gait in individuals with reduced passive ankle dorsiflexion. Gait Posture 93, 32–38. https://doi.org/10.1016/j.gaitpost.2022.01.010.

Backman, L.J., Danielson, P., 2011. Low range of ankle dorsiflexion predisposes for patellar tendinopathy in junior elite basketball players: a 1-Year prospective study. Am. J. Sports Med. 39 (12), 2626–2633. https://doi.org/10.1177/0363546511420552.

Bennell, K.L., Talbot, R.C., Wajswelner, H., Techovanich, W., Kelly, D.H., Hall, A.J., 1998. Intra-rater and inter-rater reliability of a weight-bearing lunge measure of ankle dorsiflexion. Aust. J. Physiother. 44 (3), 175–180. https://doi.org/10.1016/s0004-9514(14)60377-9.

Bridges, A.J., Holler, K.A., 2007. How many is enough? Determining optimal sample sizes for normative studies in pediatric neuropsychology. Child Neuropsychol. J. Norm. Abnorm. Dev. Child Adolesc. 13 (6), 528–538. https://doi.org/10.1080/09297040701233875.

Csapo, R., Maganaris, C.N., Seynnes, O.R., Narici, M.V., 2010. On muscle, tendon and high heels. J. Exp. Biol. 213 (Pt 15), 2582–2588. https://doi.org/10.1242/jeb.044271.

Dill, K.E., Begalle, R.L., Frank, B.S., Zinder, S.M., Padua, D.A., 2014. Altered knee and ankle kinematics during squatting in those with limited weight-bearing-lunge ankle-dorsiflexion range of motion. J. Athl. Train. 49 (6), 723–732. https://doi.org/10.4085/1062-6050-49.3.29.

Gajdosik, R.L., Vander Linden, D.W., Williams, A.K., 1999. Influence of age on length and passive elastic stiffness characteristics of the calf muscle-tendon unit of women. Phys. Ther. 79 (9), 827–838. https://doi.org/10.1093/ptj/79.9.827.

Giuliani, H.K., Gerstner, G.R., Mota, J.A., Ryan, E.D., 2019. Age-related changes in the passive properties of the plantarflexors: influence of tissue size and quality. Clin. Biomech. Bristol. Avon. 68, 53–57. https://doi.org/10.1016/j.clinbiomech.2019.05.027.

Gohil, R., Tilaye, P., 2022. Normative data for calf muscle flexibility tested by weight bearing lunge test in age group of 20-30 years - pilot study. Int. J. Health Sci. Res. 12 (2), 71–75. https://doi.org/10.52403/ijhsr.20220209.

Gosse, G., Ward, E., McIntyre, A., Banwell, H.A., 2021. The reliability and validity of the weight-bearing lunge test in a Congenital Talipes Equinovarus population (CTEV). PeerJ 9, e10253. https://doi.org/10.7717/peerj.10253.

Han, S., Lee, H., Son, S.J., Hopkins, J.T., 2023. Effect of varied dorsiflexion range of motion on landing biomechanics in chronic ankle instability. Scand. J. Med. Sci. Sports 33 (7), 1125–1134. https://doi.org/10.1111/sms.14339.

Hawkins, M.S., Storti, K.L., Richardson, C.R., et al., 2009. Objectively measured physical activity of USA adults by sex, age, and racial/ethnic groups: a cross-sectional study. Int. J. Behav. Nutr. Phys. Activ. 6, 31. https://doi.org/10.1186/1479-5868-6-31.

Hernández-Guillén, D., Tolsada-Velasco, C., Roig-Casasús, S., Costa-Moreno, E., Borja-de-Fuentes, I., Blasco, J.M., 2021. Association ankle function and balance in community-dwelling older adults. PLoS One 16 (3), e0247885. https://doi.org/10.1371/journal.pone.0247885.

Hoch, M.C., McKeon, P.O., 2011. Normative range of weight-bearing lunge test performance asymmetry in healthy adults. Man. Ther. 16 (5), 516–519. https://doi.org/10.1016/j.math.2011.02.012.

Kemoun, G., Thoumie, P., Boisson, D., Guieu, J.D., 2002. Ankle dorsiflexion delay can predict falls in the elderly. J. Rehabil. Med. 34 (6), 278–283. https://doi.org/10.1080/165019702760390374.

Koshino, Y., Takabayashi, T., Akuzawa, H., et al., 2024. Differences and relationships between weightbearing and non-weightbearing dorsiflexion range of motion in foot and ankle injuries. J. Orthop. Surg. 19 (1), 115. https://doi.org/10.1186/s13018-024-04599-x.

Malliaras, P., Cook, J.L., Kent, P., 2006. Reduced ankle dorsiflexion range may increase the risk of patellar tendon injury among volleyball players. J. Sci. Med. Sport 9 (4), 304–309. https://doi.org/10.1016/j.jsams.2006.03.015.

Marcucci, L., Reggiani, C., 2020. Increase of resting muscle stiffness, a less considered component of age-related skeletal muscle impairment. Eur. J. Transl. Myol. 30 (2), 8982. https://doi.org/10.4081/ejtm.2019.8982.

McKay, M.J., Baldwin, J.N., Ferreira, P., Simic, M., Vanicek, N., Burns, J., 2017. Normative reference values for strength and flexibility of 1,000 children and adults. Neurology 88 (1), 36–43. https://doi.org/10.1212/WNL.0000000000003466.

Menz, H.B., 2005. Analysis of paired data in physical therapy research: time to stop double-dipping? J. Orthop. Sports Phys. Ther. 35 (8), 477–478. https://doi.org/10.2519/jospt.2005.0108.

Menz, H.B., 2015. Biomechanics of the ageing foot and ankle: a mini-review. Gerontology 61 (4), 381–388. https://doi.org/10.1159/000368357.

Moisan, G., McBride, S., Isabelle, P.L., Chicoine, D., 2022. The keystone device as a clinical tool for measuring the supination resistance of the foot: a reliability study. Muscoskel. Care 20 (3), 570–576. https://doi.org/10.1002/msc.1614.

Nambiar, A., Vavachan, N., Karthikprabhu, Ravichandran H., Shetty, K.S., 2024. Normative values of weight bearing lunge test among non-athlete females - a criterion based study. RGUHS J. Physiother. 4 (1). https://doi.org/10.26463/rjpt.4_1_4.

Neumann, D.A., 2017. *Kinesiology of the Musculoskeletal System: Foundations for Rehabilitation*. Third. Elsevier.

Onate, J.A., Starkel, C., Clifton, D.R., et al., 2018. Normative functional performance values in high school athletes: the functional pre-participation evaluation project. J. Athl. Train. 53 (1), 35–42. https://doi.org/10.4085/1062-6050-458.16.

Powden, C.J., Hoch, J.M., Hoch, M.C., 2015. Reliability and minimal detectable change of the weight-bearing lunge test: a systematic review. Man. Ther. 20 (4), 524–532. https://doi.org/10.1016/j.math.2015.01.004.

Protopapadaki, A., Drechsler, W.I., Cramp, M.C., Coutts, F.J., Scott, O.M., 2007. Hip, knee, ankle kinematics and kinetics during stair ascent and descent in healthy young individuals. Clin. Biomech. Bristol. Avon. 22 (2), 203–210. https://doi.org/10.1016/j.clinbiomech.2006.09.010.

Ralphs, J.R., Benjamin, M., 1994. The joint capsule: structure, composition, ageing and disease. J. Anat. 184 (Pt 3), 503–509.

Rao, Y., Yang, N., Gao, T., et al., 2024. Effects of peak ankle dorsiflexion angle on lower extremity biomechanics and pelvic motion during walking and jogging. Front. Neurol. 14, 1269061. https://doi.org/10.3389/fneur.2023.1269061.

Redmond, A.C., Crosbie, J., Ouvrier, R.A., 2006. Development and validation of a novel rating system for scoring standing foot posture: the foot posture index. Clin. Biomech. 21 (1), 89–98.

Riddle, D.L., Pulisic, M., Pidcoe, P., Johnson, R.E., 2003. Risk factors for plantar fasciitis: a matched case-control study. J. Bone Jt. Surg. Am. 85 (5), 872–877.

Roddy, E., Thomas, M.J., Marshall, M., et al., 2015. The population prevalence of symptomatic radiographic foot osteoarthritis in community-dwelling older adults: cross-sectional findings from the clinical assessment study of the foot. Ann. Rheum. Dis. 74 (1), 156–163. https://doi.org/10.1136/annrheumdis-2013-203804.

Roman-Blas, J.A., Castañeda, S., Largo, R., Herrero-Beaumont, G., 2009. Osteoarthritis associated with estrogen deficiency. Arthritis Res. Ther. 11 (5), 241. https://doi.org/10.1186/ar2791.

Schober, P., Boer, C., Schwarte, L.A., 2018. Correlation coefficients: appropriate use and interpretation. Anesth. Analg. 126 (5), 1763. https://doi.org/10.1213/ANE.0000000000002864.

Simondson, D., Brock, K., Cotton, S., 2012. Reliability and smallest real difference of the ankle lunge test post ankle fracture. Man. Ther. 17 (1), 34–38. https://doi.org/10.1016/j.math.2011.08.004.

Simondson, J., Simondson, D., Formby, C., Brock, K., 2022. The ankle lunge test for measurement of dorsiflexion in individuals following stroke; reliability, validity and evaluation of change following intervention for spasticity of the lower limb. Physiother. Res. Int. J. Res. Clin. Phys. Ther. 27 (1), e1925. https://doi.org/10.1002/pri.1925.

Smith, M.D., Lee, D., Russell, T., Matthews, M., MacDonald, D., Vicenzino, B., 2019. How much does the talocrural joint contribute to ankle dorsiflexion range of motion during the weight-bearing lunge test? A cross-sectional radiographic validity study. J. Orthop. Sports Phys. Ther. 49 (12), 934–941. https://doi.org/10.2519/jospt.2019.8697.

Soucie, J.M., Wang, C., Forsyth, A., et al., 2011. Range of motion measurements: reference values and a database for comparison studies. Haemoph. Off J. World Fed. Hemoph. 17 (3), 500–507. https://doi.org/10.1111/j.1365-2516.2010.02399.x.

Vandervoort, A.A., Chesworth, B.M., Cunningham, D.A., Paterson, D.H., Rechnitzer, P.A., Koval, J.J., 1992. Age and sex effects on mobility of the human ankle. J. Gerontol. 47 (1), M17–M21. https://doi.org/10.1093/geronj/47.1.M17.

Willems, T.M., Witvrouw, E., Delbaere, K., Mahieu, N., De Bourdeaudhuij, L., De Clercq, D., 2005. Intrinsic risk factors for inversion ankle sprains in Male subjects: a prospective study. Am. J. Sports Med. 33 (3), 415–423. https://doi.org/10.1177/0363546504268137.